

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 10%

QUESTIONS: 30

Total Marks:30

Annexure A

MCQ TEST

Criteria	Conceptual Understanding (3)	Accuracy of Answers (2.5)	Application of Knowledge (2)	Time Management (1.5)	Question Interpretation (1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q						

Code of Conduct:

I G Madhulatha(192311295) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

G Madhulatha

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANNEXURE A

MCQ Test:

1. Consider a simple linear regression model with one independent variable. Which of the following is true? (CO3)

Option_a: The regression line always passes through the mean of X and Y.

Option_b: The slope is always positive.

Option_c: The intercept is always positive.

Option_d: The correlation coefficient is always 1.

2. Which of the following statements is correct about Decision Trees? (CO3)

Option_a: Decision trees can only handle numerical data.

Option_b: Decision trees are always prone to overfitting.

Option_c: Decision trees can be used for classification and regression.

Option_d: None of the above.

3. Which of the following methods can be used for pruning in decision trees? (CO3)

Option_a: Pre-pruning

Option_b: Post-pruning

Option_c: Both (a) and (b)

Option_d: None of the above.

4. Which of the following is a disadvantage of Decision Trees? (CO3)

Option_a: Prone to overfitting.

Option_b: Requires large datasets.

Option_c: Cannot handle categorical data.

Option_d: Always has a high accuracy.

5. What does Bayes' Theorem primarily help with in probability and statistics? (CO3)

Option_a: Determining prior probabilities only

Option_b: Updating probabilities based on new evidence

Option_c: Computing standard deviation of a dataset

Option_d: Finding mean and median in distributions

6. In Bayes' Theorem, what does the term $P(A|B)$ represent? (CO3)

Option_a: The probability of event B given event A has occurred

Option_b: The probability of event A given event B has occurred

Option_c: The joint probability of A and B

Option_d: The marginal probability of B

7. A factory has two machines, A and B. Machine A produces 60% of the products, and machine B produces 40%. If 2% of A's products and 3% of B's products are defective, what is the probability that a randomly selected defective product came from Machine B? (CO3)

Option_a: 0.40

Option_b: 0.50

Option_c: 0.60

Option_d: 0.75

8. Which of the following is not required to apply Bayes' Theorem? (CO3)

Option_a: Prior probability of the event

Option_b: Conditional probability of the evidence

Option_c: Total number of possible outcomes

Option_d: Probability of the evidence

9. A doctor uses a test to diagnose a disease that affects 1% of a population. The test correctly

identifies the disease 95% of the time but also gives a false positive in 5% of cases. If a patient tests positive, what is the probability they actually have the disease? (CO3)

Option_a: 0.16

Option_b: 0.34

Option_c: 0.95

Option_d: 0.50

10. Bayes' Theorem is most useful when dealing with which type of probability? (CO3)

Option_a: Independent events

Option_b: Conditional probability

Option_c: Mutually exclusive events

Option_d: Uniform probability distribution

11. If a hypothesis HHH has a prior probability of 0.3 and new evidence EEE has a likelihood of 0.8 given HHH, how does Bayes' Theorem adjust the probability of HHH after seeing EEE? (CO3)

Option_a: The probability of HHH increases

Option_b: The probability of HHH decreases

Option_c: The probability of HHH remains unchanged

Option_d: The probability of HHH is completely unknown

12. A bag contains 5 red and 7 blue balls. A ball is drawn randomly and found to be red. What is the probability that the next ball drawn is also red (assuming the first ball is not replaced)? (CO3)

Option_a: 5/12

Option_b: 4/11

Option_c: 5/11

Option_d: 6/12

13. Which field heavily relies on Bayes' Theorem for updating beliefs based on new data? (CO3)

Question 230: END

Option_a: Machine Learning

Option_b: Geometry

Option_c: Classical Physics

Option_d: Cryptography

14. Why is backpropagation crucial in training deep neural networks? (CO3)

Option_a: It prevents overfitting by pruning unnecessary weights.

Option_b: It updates weights using the gradient of the loss function.

Option_c: It ensures each layer has the same activation values.

Option_d: It removes redundant layers to optimize the model.

15. During backpropagation, what happens when the learning rate is too high? (CO3)

Option_a: The model converges faster with high accuracy.

Option_b: The model may overshoot the optimal weight values and fail to converge.

Option_c: The model will always get stuck in local minima.

Option_d: The number of hidden layers increases automatically.

16. What is the main advantage of rule-based induction models? (CO3)

Question 247: END

Option_a: They provide an interpretable set of rules for decision-making.

Option_b: They work well with both numerical and categorical data.

Option_c: They can generalize well without requiring large datasets.

Option_d: Both (a) and (b)

17. Which of the following is an example of a rule-based induction algorithm? (CO3)

Question 248: END

Option_a: Decision Tree

Option_b: Naïve Bayes

Option_c: Support Vector Machine

Option_d: K-Nearest Neighbors

18. In rule-based induction, what happens when the rules are too specific to the training data?

Option_a: The model may suffer from overfitting.

Option_b: The model will generalize better.

Option_c: The rules will be shorter and more efficient.

Option_d: The accuracy will always increase.

19. Which approach is commonly used to simplify rule-based induction models? (CO3)

Question 250: END

Option_a: Pruning redundant or less significant rules

Option_b: Adding more features to the dataset

Option_c: Increasing the depth of the decision tree

Option_d: Using more training samples

20. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)

Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions

Option_b: To calculate the computational complexity of the classifier

Option_c: To determine the training time of the classifier

Option_d: To identify the features used by the classifier

21. Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)

Question 252: END

Option_a: Precision

Option_b: Recall

Option_c: Accuracy

Option_d: F1-Score

22. In a binary classification problem, what does the True Positive (TP) value represent? (CO3)

Question 253: END

Option_a: The number of negative instances correctly classified as negative

Option_b: The number of positive instances correctly classified as positive

Option_c: The number of positive instances incorrectly classified as negative

Option_d: The number of negative instances incorrectly classified as positive

23. What is the formula for calculating precision in a classification problem? (CO3)

Option_a: Precision = $TP / (TP + FP)$

Option_b: Precision = $TP / (TP + FN)$

Option_c: Precision = $(TP + TN) / (TP + TN + FP + FN)$

Option_d: Precision = $TP / (FP + FN)$

24. Which of the following metrics is most useful when the classes are imbalanced? (CO3)

Option_a: Accuracy

Option_b: F1-Score

Option_c: Recall

Option_d: Specificity

25. What does a high recall value indicate in a classification model? (CO3)

Question 256: END

Option_a: The model has a low number of false positives

Option_b: The model has a low number of false negatives

Option_c: The model has a high number of true negatives

Option_d: The model has a high number of false positives

Correct_option: The model has a low number of false negatives

26. Which of the following is an example of a rule-based induction algorithm? (CO3)

Option_a: Decision Tree

Option_b: Naïve Bayes

Option_c: Support Vector Machine

Option_d: K-Nearest Neighbor

27. In rule-based induction, what happens when the rules are too specific to the training data? (CO3)

Option_a: The model may suffer from overfitting.

Option_b: The model will generalize better.

Option_c: The rules will be shorter and more efficient.

Option_d: The accuracy will always increase.

28. Which approach is commonly used to simplify rule-based induction models? (CO3)

Option_a: Pruning redundant or less significant rules

Option_b: Adding more features to the dataset

Option_c: Increasing the depth of the decision tree

Option_d: Using more training samples

29. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)

Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions

Option_b: To calculate the computational complexity of the classifier

Option_c: To determine the training time of the classifier

Option_d: To identify the features used by the classifier

30. Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)

Question 252: END

Option_a: Precision

Option_b: Recall

Option_c: Accuracy

Option_d: F1-Score

ANSWERS

1. A – Regression line passes through the mean values of X and Y.
2. C – Decision Trees are used for both classification and regression.
3. C – Both pre-pruning and post-pruning reduce overfitting.
4. A – Decision Trees can easily overfit the training data.
5. B – Bayes' Theorem updates probability using new evidence.
6. B – $P(A|B)$ means probability of A given B.
7. B – Using Bayes' Theorem, probability = 0.50.
8. C – Total possible outcomes are not directly required.
9. A – Actual disease probability after a positive test ≈ 0.16 .
10. B – Bayes' Theorem mainly deals with conditional probability.
11. A – Strong evidence increases the probability of the hypothesis.
12. B – After one red is removed, 4 red remain out of 11 balls: 4/11.
13. A – Machine Learning uses Bayes' Theorem to update predictions.
14. B – Backpropagation updates weights using loss gradients.
15. B – A high learning rate can overshoot the optimal solution.
16. D – Rule-based models are interpretable and can handle numerical/categorical data.
17. A – Decision Trees can generate IF–THEN rules.
18. A – Very specific rules cause overfitting.
19. A – Pruning removes unnecessary or less important rules.
20. A – A confusion matrix shows correct and incorrect predictions.
21. C – Accuracy measures correctly classified instances out of total.
22. B – True Positive means a positive case correctly predicted as positive.
23. A – Precision = $TP / (TP + FP)$.
24. B – F1-Score is useful for imbalanced datasets.
25. B – High recall means fewer false negatives.
26. A – Decision Tree is an example of rule-based induction.
27. A – Too-specific rules lead to overfitting.
28. A – Pruning simplifies a rule-based model.
29. A – Confusion matrix evaluates correct and incorrect classifications.
30. C – Accuracy = proportion of correct predictions out of total.